de Bruijn

binary

[0x3a] alphabet size $k = 2 \equiv \{0,1\}$

subseq length $n = 6$ $\left(2^6 = 64 \text{ possible positions}\right)$

$\alpha = B(2,6)$

cyclically rotating the de Bruijn seq by p positions

$n\,b = 2^p \cdot \alpha \pmod{2^{64}}$

negate (2's-compl) ie $rdx = 0 - rdx$

position of lowest set bit via de Bruijn

```
mov     rdx, rax
neg     rdx
and     rax, rdx
mov     rdx, 0x218a392cd3d5dbf
mul     rdx
shr     rax, 0x3a
xlatb
```

$rdx := rax_0$

$rdx = -rax_0$

$rax = rax_0 \wedge (-rax_0)$

$=: \alpha$

$\Rightarrow rax = (rax_0 \wedge (-rax_0)) \cdot \alpha$

$rax = ((rax_0 \wedge (-rax_0)) \cdot \alpha) >> 0x3a$ ← offset of lookup table

$rax \ni al := ds:[ebx + al]$ (unsigned; ebx: root/base + offset)

will end up inside

value of lowest set bit

$rax = ((x \wedge -x)\alpha) >> 0x3a$

ie $rax \leq max(al)$

$0x40 - 0x3a = 6$ bits (most-sig)

$\left(\left(\text{value of lowest set bit}\right) \cdot \alpha\right) >> 0x3a$ (const value)

$index = ((2^p \cdot B(2,6)) \pmod{2^{64}}) >> 0x3a$

$rax \ni al := ds:[ebx + al]$ (const; [0x3a])

$= p$ (desired bit position)

al_{final}